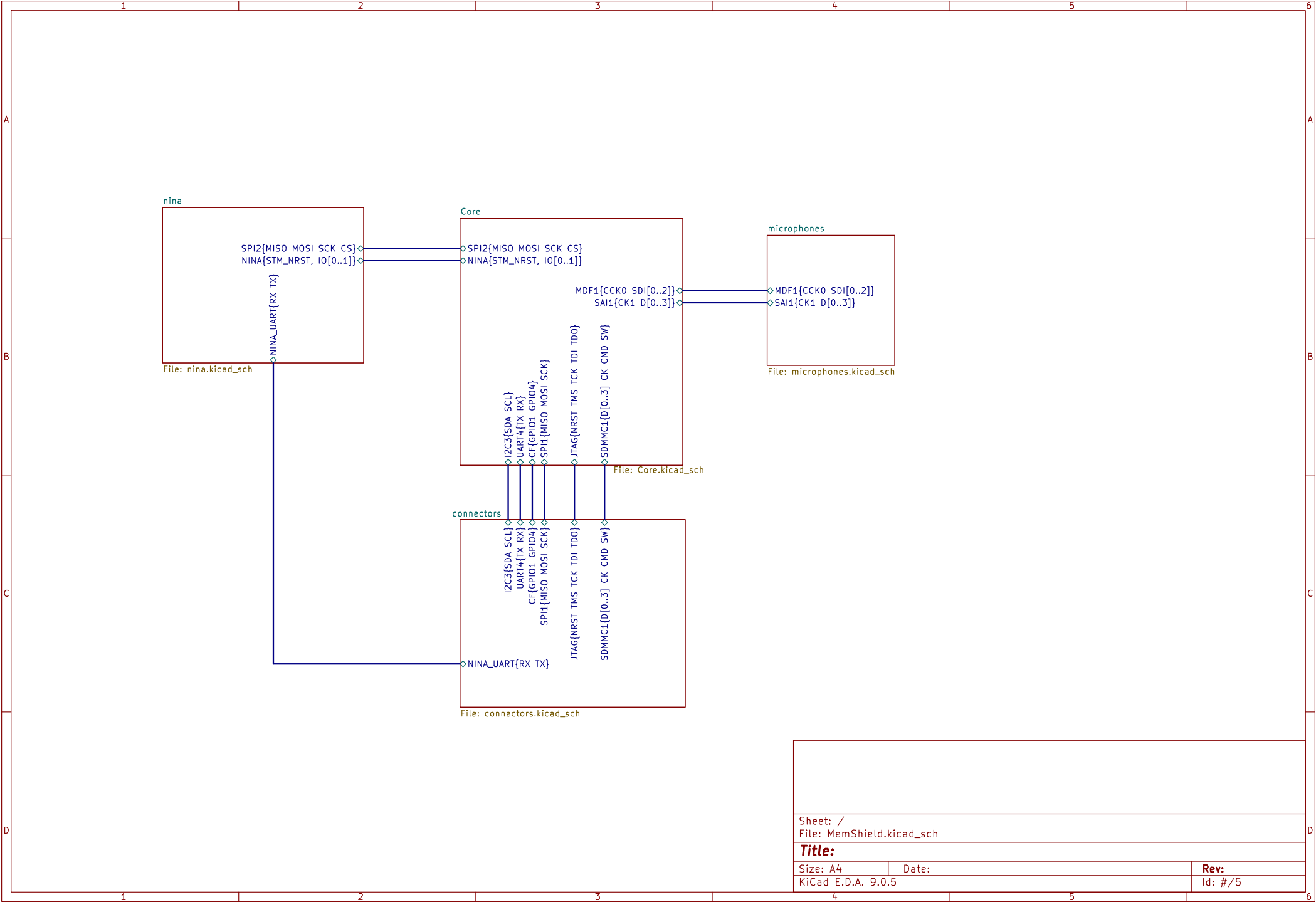




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The schematic diagram illustrates a TP563001DRCR DC-DC converter circuit. The input voltage is VBat, which is connected to the VIN pin (pin 5) of the converter. A 10uF capacitor (C2) is connected between VBat and GND. The output of the converter is VCore, which is connected to the VOUT pin (pin 1). A 22uF capacitor (C14) is connected between VCore and GND. The converter is powered by VBat and outputs VCore. The circuit is connected to a GND1 and GND2 ground plane.

VCORE

100nF C3

100nF C6

100nF C8

10uF C9

100nF C10

100nF C11

1uF C13

GND

SDMMC1{D[0..3] CK CMD SW} SDMMC1.D0
 SDMMC1.D1
 SDMMC1.D2
 SDMMC1.D3
 SDMMC1.CK
 SDMMC1.CMD
 SDMMC1.SW

The diagram shows the pin connections for the STM32U575RG6 microcontroller. The pins are organized into two columns: pins 1-15 on the left and pins 16-36 on the right. The connections are as follows:

- Pin 1:** VBAT
- Pin 2:** VDD
- Pin 3:** VDD
- Pin 4:** VDD
- Pin 5:** VDD
- Pin 6:** VDD
- Pin 7:** VDDA
- Pin 8:** VDDUSB
- Pin 9:** JTAG.NRST
- Pin 10:** NRST
- Pin 11:** PH0
- Pin 12:** PH1
- Pin 13:** PH3
- Pin 14:** PA0
- Pin 15:** PA1
- Pin 16:** PA2
- Pin 17:** PA3
- Pin 18:** PA4
- Pin 19:** PA5
- Pin 20:** PA6
- Pin 21:** PA7
- Pin 22:** PA8
- Pin 23:** PA9
- Pin 24:** PA10
- Pin 25:** PA11
- Pin 26:** PA12
- Pin 27:** PA13
- Pin 28:** PA14
- Pin 29:** PA15
- Pin 30:** PC0
- Pin 31:** PC1
- Pin 32:** PC2
- Pin 33:** PC3
- Pin 34:** PC4
- Pin 35:** PC5
- Pin 36:** PC6
- Pin 37:** PC7
- Pin 38:** PC8
- Pin 39:** PC9
- Pin 40:** PC10
- Pin 41:** PC11
- Pin 42:** PC12
- Pin 43:** PC13
- Pin 44:** PC14
- Pin 45:** PC15
- Pin 46:** VCAP
- Pin 47:** VSS
- Pin 48:** VSSA

Additional connections shown include:

- JTAG.NRST** connected to pin 9.
- RCC_OSC_IN** connected to pin 5.
- RCC_OSC_OUT** connected to pin 6.
- GND** connected to pin 60.
- SDMMC1.CMD** connected to pin 54.
- I2C3.SDA** connected to pin 8.
- SPI2.MISO** connected to pin 10.
- SAI1.D1** connected to pin 11.
- SAI1.D3** connected to pin 25.
- SPI2.CS** connected to pin 37.
- NINA.IO1** connected to pin 38.
- SDMMC1.D0** connected to pin 39.
- SDMMC1.D2** connected to pin 51.
- SDMMC1.D3** connected to pin 52.
- SDMMC1.CK** connected to pin 53.
- C7 4.7uF** connected to pin 30.
- GND** connected to pin 30.
- GND** connected to pin 18.
- U3 STM32U575RG6** connected to pin 18.
- UART4.TX** connected to pin 14.
- UART4.RX** connected to pin 15.
- SAI1.CK1** connected to pin 17.
- CF.GPIO1** connected to pin 20.
- SPI1.SCK** connected to pin 21.
- SPI1.MISO** connected to pin 22.
- I2C3.SCL** connected to pin 23.
- CF.GPIO4** connected to pin 41.
- NINA.STM_NRST** connected to pin 42.
- NINA.IO0** connected to pin 43.
- FAULT_LED** connected to pin 44.
- SPI1.MOSI** connected to pin 45.
- JTAG.TMS** connected to pin 46.
- JTAG.TCK** connected to pin 49.
- JTAG.TDI** connected to pin 50.
- AUDIOCLK_SIG** connected to pin 26.
- MDF1.SDIO** connected to pin 27.
- JTAG.TDO** connected to pin 28.
- SDMMC1.SW** connected to pin 56.
- MDF1.CCK0** connected to pin 61.
- SAI1.D2** connected to pin 62.
- MDF1.SDIO1** connected to pin 29.
- SPI2.SCK** connected to pin 34.
- MDF1.SDIO2** connected to pin 35.
- SPI2.MOSI** connected to pin 36.

SAI1{CK1 D[0..3]} $\diamond$ SAI1.CK1
SAI1.D1
SAI1.D2
SAI1.D3

NINA{STM_NRST, IO[0..1]} $\diamond$ NINA.STM_NRST
NINA.IO0
NINA.IO1

CF{GPIO1 GPIO4} $\diamond$ CF.GPIO1
CF.GPIO4

I2C3{SDA SCL} $\diamond$ I2C3.SDA
I2C3.SCL

UART4{TX RX} $\diamond$ UART4.TX
UART4.RX

SPI1{MISO MOSI SCK} $\diamond$ SPI1.MISO
SPI1.MOSI
SPI1.SCK

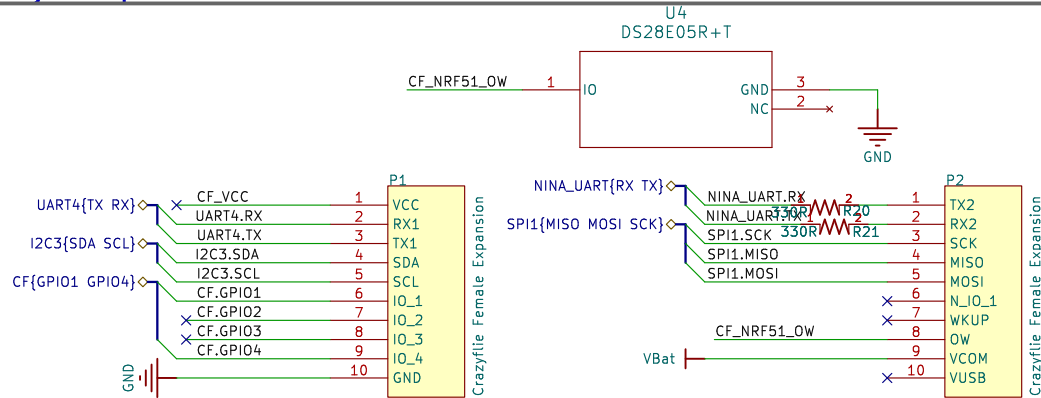
SPI2{MISO MOSI SCK CS} $\diamond$ SPI2.MISO
SPI2.MOSI
SPI2.SCK
SPI2.CS

Diagram D8 shows a circuit for a GREEN LED. It consists of a 3V3 voltage source connected in series with a 47k resistor (R4) and the GREEN LED. The LED is connected to ground (GND) through a 100nF capacitor and a 10k resistor in parallel. The LED is labeled 'GREEN LED' and the resistor is labeled 'R4 47k'.

Diagram D9 shows a circuit for a RED LED. It consists of a FAULT_LED voltage source connected in series with a 47k resistor (R7) and the RED LED. The LED is connected to ground (GND) through a 100nF capacitor and a 10k resistor in parallel. The LED is labeled 'RED LED' and the resistor is labeled 'R7 47k'.

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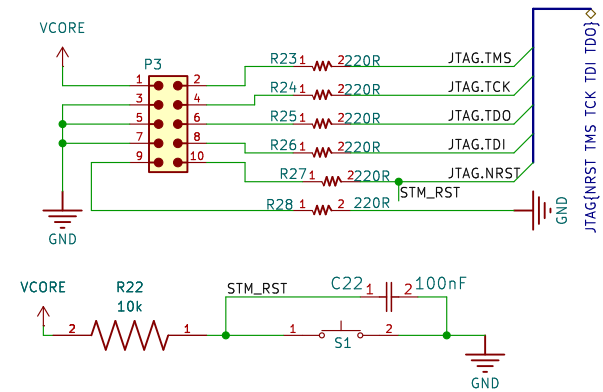
Crazyflie Expansions



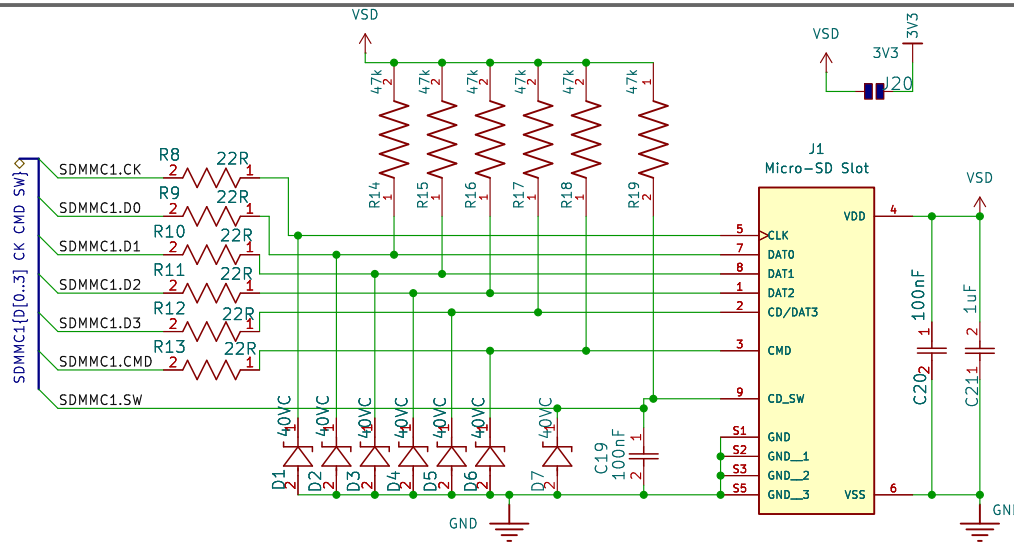
Please note the following about the Crazyflie expansion connectors:

- * VCOM is power directly from the battery/USB and is not regulated (max 1A)
- * VUSB is connected directly to the USB which means this will supply at 4.5-5.5V when the USB is connected and can be used for charging the battery when the
- * VCC is regulated to 3V0 (max. 100mA)

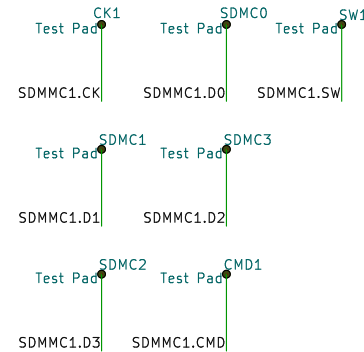
JTAG



SD Card Slot



Signal Test Pads



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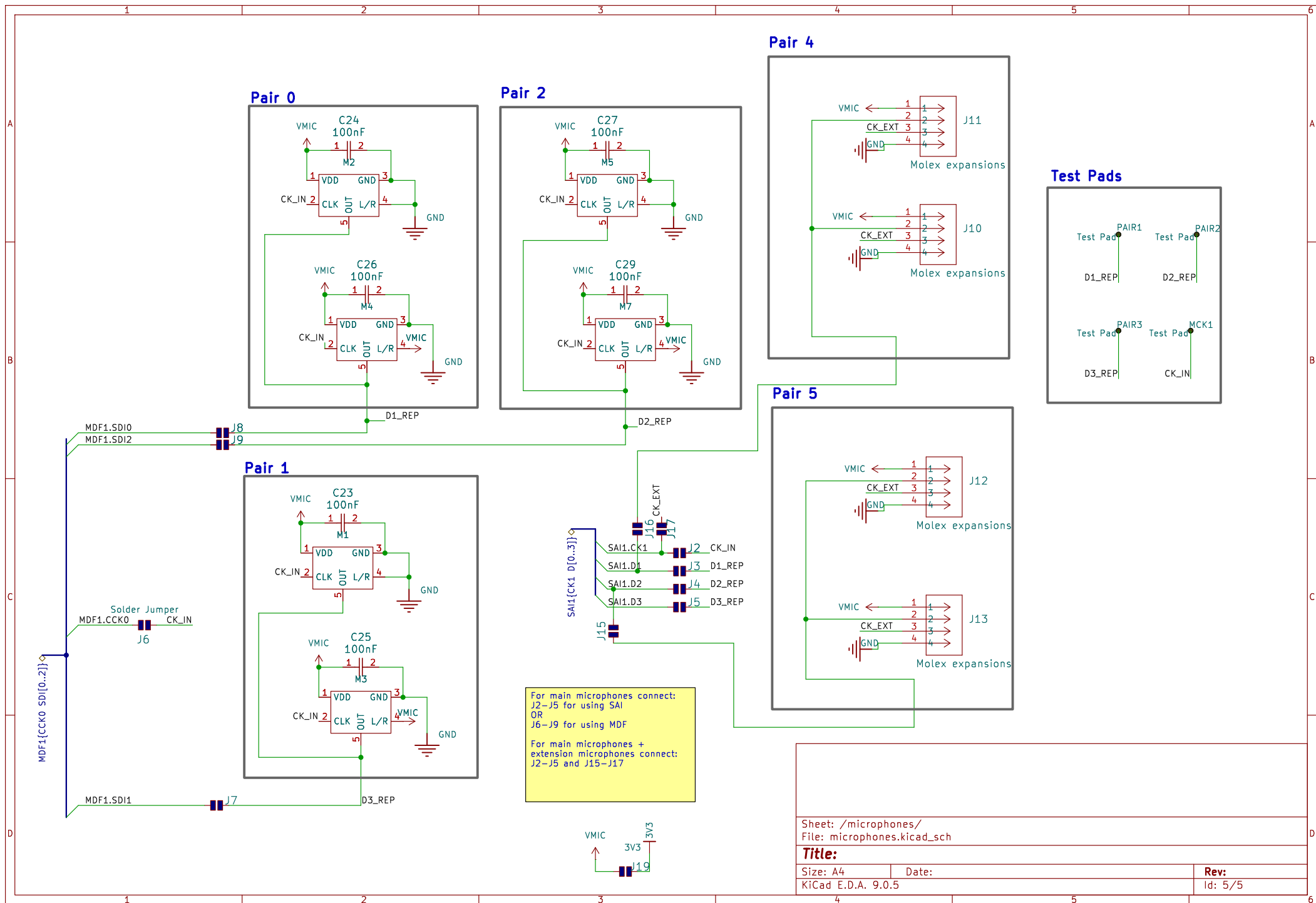
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The diagram illustrates the JTAG interface for the VN1A. The 5-pin connector P5 is connected to the JTAG pins of the VN1A. The pins are: 1 (VN1A_JTAG_TMS), 2 (VN1A_RST), 3 (VN1A_NRST, with a 220Ω resistor), 4 (VN1A_JTAG_TCK), 5 (VN1A_JTAG_TDI), 6 (VN1A_JTAG_TDO), 7 (GND), and 8 (GND). A separate circuit shows the VN1A_RST pin connected to a 100nF capacitor, which is connected to GND. A solder jumper J18 is shown connected to the VN1A_RST pin.

[illegible]

Diagram illustrating the connection of SPI2 pins to NINA module pins:

- `SPI2{MISO MOSI SCK CS}` is connected to:
 - `NINA.STM_NRST`
 - `NINA.IO0`
 - `NINA.IO1`
 - `NINA.UART{RX TX}`

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